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SOP310

①

a) BFS

J, F, C, P, D, G, L, S, $\textcircled{V} \leftarrow \text{Goal}$

b) DFS

J, F, D, N, G, C, P, L, S, Q, U, $\textcircled{V} \leftarrow \text{Goal}$

②

List: 4, 6, 1, 3, 9, 2, 7, 8, 5

a) branching factor = 8

Since there are 9 numbers, there are 8 adjacent pairs that can be swapped. What this says is at every state there are always 8 potential actions to be taken.

b) An admissible heuristic could be the number of elements that are out of order.